Haris Amir



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Machine Learning Engineer with 3+ years of experience developing **3D perception, autonomous-driving vision, RGB-D reconstruction, multimodal perception, and deep-learning systems**. Experienced with **PyTorch, point clouds, SLAM, ROS 2, Vision Transformers, real-time inference, and distributed GPU training**, with production-oriented ML experience using **Docker, Azure ML, SLURM, HPC, W&B, and CI/CD**.

PROFESSIONAL EXPERIENCE

Volkswagen Group Jan 2026 – Oct 2026 | Wolfsburg, Germany

Internship / Master Thesis

- Conducted research on efficient **Vision Transformer architectures for multi-view autonomous-driving perception**, reducing computational complexity by ~**50%** while maintaining <**2% accuracy degradation**.
- Worked extensively with **multi-camera geometry and calibration**, including **camera intrinsics/extrinsics, projection matrices, coordinate-frame transformations, distortion correction, and vehicle-frame 3D perception representations**.
- Applied **epipolar geometry, homographies, triangulation, and image-to-world geometric transformations** for reasoning across calibrated camera views.
- Worked with **BEV transformations** and camera-to-vehicle coordinate mappings for spatial representation of multi-camera perception outputs.
- Implemented and evaluated model variants in **PyTorch** using systematic **benchmarking, ablation studies, and large-scale GPU experiments**.
- Designed reproducible distributed training and experimentation workflows using **Azure ML, SLURM, HPC, GPU clusters, and cloud storage**.

Deutsches Forschungszentrum für Künstliche Intelligenz - DFKI Aug 2024 – Dec 2025

Research Assistant - AI Kaiserslautern, Germany

- Developed and containerized an end-to-end **computer-vision defect detection system** using **Python, PyTorch, RTMDet, and Docker** for manufacturing quality-control applications.
- Designed a **multimodal 3D perception framework** integrating **3D object detection, SLAM-based scene understanding, language grounding, and image captioning** into a unified spatial-reasoning pipeline.
- Worked with **3D spatial representations, point-cloud-based perception, coordinate transformations, and camera/sensor geometry** for multimodal scene understanding.
- Integrated and evaluated multiple ML and perception components, performing **technical benchmarking, experimentation, and documentation** for research and application-oriented stakeholders.

TenX Dec 2022 – Feb 2024 | Remote

Data Scientist

- Developed a machine learning algorithm (achieving **88%** F1 score) for the quantification of total viable bacteria (TVB)from **microscopic images** of swab samples and poultry rinsate. Reduced result turnaround from **24–48** hours tominutes, replacing conventional lab-based methods with rapid AI inference.
- Improved accuracy of existing cell detection models by up to **3%** through integrating attention gates with the **U-Net** architecture and leveraging synthetic data generation for model robustness.
- Built an **ML Ops** pipeline using **Weights & Biases (W&B)** to track datasets, model weights, and hyperparameters. Automated model training and evaluation reports with Jobs and Launches, reducing end-to-end workflow time by **50%**.
- Developed and maintained production infrastructure using **Azure** enabling **training and deployment** of models at scale, significantly improving operational efficiency and system reliability.

PROJECTS

Continual Event-Based Object Detection for Autonomous Driving

- Developed a **class-incremental** continual-learning detector for event-camera data (**1 Mpx automotive dataset**) using a Recurrent Vision Transformer **(RVT)** backbone and **SDCoT**-inspired static-teacher pseudo-labeling and distillation.
- Achieved **27.3% overall mAP@50** (using supervised+distillation losses) vs. where naive fine-tuning results collapsed to almost **0,** demonstrating strong mitigation of catastrophic forgetting in continual learning setting for event based data.
- Preserved real time performance (**<12 ms** inference on NVIDIA T4).

RGB-D Scene Reconstruction and 3D Point Cloud Generation using ROS 2

- Developed a **ROS 2** pipeline to reconstruct colourised **3D point clouds** from synchronised **RGB** and depth streams recorded with a **ZED** camera.
- **Back-projected** depth images into 3D using the pinhole camera model and camera intrinsic parameters.
- Implemented timestamp-based RGB-depth synchronisation, invalid-depth filtering, point-cloud downsampling, multi-frame accumulation, and **PLY** export.
- Reconstructed indoor environments from real RGB-D sensor data

PUBLICATIONS

Efficient Vision Transformer-based Multi-View 3D Perception, Aug 2026
WACV 2027

- Master Thesis - Submitted

ReConText3D: Replay-based Continual Text-to-3D Generation, Jun 2026
CVPR 2026
Denver, Colorado, USA

- **First continual learning framework for text-to-3D generation**, enabling generative models to learn new 3D concepts while mitigating catastrophic forgetting.
- Proposed a replay-based learning strategy and introduced the **Toys4K-CL benchmark** for continual **3D generation**.

EDUCATION

Master's Degree, Computer Science 2024 – 2026 | Kaiserslautern, Germany
Technische Universität Kaiserslautern (RPTU)

- **Major**: Intelligent Systems, **Minor**: Software Engineering
- **Master's Thesis:** Increasing efficiency of End-to-End Perception Models (Volkswagen Group, **WACV '27**)

Bachelor's Degree, Software Engineering 2018 – 2022 | Islamabad, Pakistan
National University of Sciences and Technology (NUST)

SKILLS

Programming: Python, SQL, JavaScript, FastAPI, REST APIs, NumPy, Pandas

3D Vision & Robotics: 3D Perception• LiDAR • SLAM • ROS 2 • Point Clouds • KITTI-360 • Computer Vision

ML: PyTorch, Transformers, Computer Vision, Multimodal AI

MLOps & Infrastructure: Docker, Azure ML, AWS, SLURM, HPC, GPU Computing, CI/CD, GitHub Actions, Weights & Biases, Grafana